CS110 / INDEPENDENCE ROAD

team / Rodolfo Sierra-Trujillo

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Record: FORM-51426BF1736A124E5873680F

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SHA-256: 459f3a28c47bf4e37a2ac1426e639b30fd8d66894e2067cbd79199df8865046f

Verify the record number at courses.addp2p.com/development-road/independence/index.html?view=verify

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team

TEAM-0E21D65CAEA6737ECD450722

Name

Starfighter

lead

Rodolfo Sierra-Trujillo

Team members

0

Name

Rodolfo Sierra-Trujillo

verificationCode

VIEW-E271AFC6D0D551F310A7C58D

Preparation record

PREP-5924A7036BB07FBF208B0CCB

Contribution record

FORM-E8BEB08EE808039F84FB7550

timeA

Time code

TIME-720C21931411A53D619E92E4

Name

Rodolfo Sierra-Trujillo

Action

A

Server arrival

2026-11-10T21:01:50.364Z

Class starts

2026-11-10T20:45:00Z

Calculated late points

15

Late deduction points

15

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Review status

Calculated awaiting instructor review

Attention Accord plan

mission

Since I am working solo, I decided that my mission will be to create a simple 3D space shooter game using Unity and C#. The story will be about a spaceship pilot who has to survive waves of enemy ships while traveling through space. My main playable goal is to let the player control a spaceship, move around the screen, and shoot at enemies to earn points. I want to start with basic movement and shooting before adding more features. I will use Visual Studio to write my C# scripts and GitHub to save and track my progress. My goal is to create a working game that is fun to play while improving my programming and game development skills.

framework

Since I am working solo, I decided to use the ADD (Attribute-Driven Design) framework to guide my decisions while creating my 3D space shooter game in Unity. I chose this framework because it will help me focus on the most important parts of my game before adding more complicated features. My first priority will be creating a 3D spaceship that the player can control and use to shoot enemies. After that, I want to add enemy spaceships, collisions, a health system, and a score system. I will test each feature before moving on to the next one. Using this framework will help me stay organized, improve my C# programming skills, and make sure my game is playable and completed on time.

pipeline

Story goal owner Me (solo) 2026-11-10T12:45 to 2026-11-10T13:00 Pacific

Inputs / assets owner Me (solo) 2026-11-10T13:00 to 2026-11-10T13:30 Pacific

Build + configure owner Me (solo) 2026-11-10T13:30 to 2026-11-10T16:30 Pacific

Output / scene owner Me (solo) 2026-11-10T16:30 to 2026-11-10T17:30 Pacific

Test + handoff owner Me (solo) 2026-11-10T17:30 to 2026-11-10T19:30 Pacific

My production pipeline shows how I plan to create my 3D space shooter game using Unity and C#. Since I am working solo, I will be responsible for every stage of development. My goal is to create a playable game where the player controls a spaceship and shoots enemy ships. I will use Unity, Visual Studio, C#, and 3D models to build my project. My main tasks will be creating spaceship movement, shooting mechanics, enemies, and collisions. After completing these features, I will test the game to make sure everything works correctly. I will also use GitHub to save my progress. This plan will help me stay organized and complete my project one step at a time.

tech

Since I am working solo, I decided to use tools that will help me create my 3D space shooter game and keep everything organized.

1. **Unity (Game Engine):** I chose Unity because it supports 3D game development and gives me the tools to create spaceships, enemies, and environments.

2. **C# (Programming Language):** I chose C# because it works with Unity and allows me to program spaceship movement, shooting, enemy behavior, and collisions.

3. **Visual Studio (Code Editor):** I will use Visual Studio to write my C# scripts, find errors, and improve my code.

4. **Unity Packages:** I plan to use Unity's Input System for player controls and the built-in 3D physics system for

detecting collisions.

5. **3D Assets:** I will start with basic 3D shapes for my spaceship and enemies. Later, I can add more detailed models, laser effects, and a space background.

6. **Git and GitHub (Version Control):** I will use Git and GitHub to save my progress, track changes, and make my project accessible to my professor.

Since I am working solo, I will be responsible for checking that all my tools, packages, and assets work correctly. I will test my C# scripts in Unity, check that my 3D models display properly, and make sure my GitHub repository stays updated. I chose these tools because they will help me build a playable 3D game while improving my programming skills.

Time code

Classes & components

Class | Responsibility | GameObject

PlayerController | MonoBehaviour, Rigidbody | Controls spaceship movement

PlayerShooting | MonoBehaviour | Fires laser projectiles

Laser | MonoBehaviour, Collider | Moves lasers and detects hits

EnemyController | MonoBehaviour, Collider | Controls enemy behavior

GameManager | MonoBehaviour | Tracks score and game state

PlayerHealth | MonoBehaviour | Tracks player health and damage

Variables / properties

Name : type | Value / purpose | Used by

moveSpeed | float | Spaceship movement speed

laserSpeed | float | Laser projectile speed

health | int | Player health

score | int | Player's points

damage | int | Damage from lasers

laserPrefab | GameObject | Laser object to spawn

f(x) Functions

Function | Inputs result | When called

MovePlayer() | PlayerController | Moves the spaceship

Shoot() | PlayerShooting | Creates laser projectiles

TakeDamage() | PlayerHealth | Reduces player health

OnTriggerEnter() | Laser | Detects laser collisions

AddScore() | GameManager | Adds points for destroyed enemies

Packages / using

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Package / namespace | Why included | Tested

using UnityEngine; | Accesses Unity classes such as MonoBehaviour, Rigidbody, and GameObject | [ ]

using UnityEngine.InputSystem; | Allows me to use Unity's newer Input System for spaceship controls | [ ]

using UnityEngine.UI; | Allows me to create UI elements such as health bars | [ ]

using TMPro; | Allows me to display score and other text using TextMeshPro | [ ]

Unity Input System package | Handles keyboard and controller input | [ ]

Unity 3D Physics | Handles collisions and Rigidbody physics | [ ]

communication

Team check-ins

Description / contact / owner | When Pacific | Agreed

Review my Unity project goals | 2026-11-11T19:21 | [x]

Check my C# code and progress | 2026-11-12T19:22 | [x]

Test spaceship movement and shooting | 2026-11-13T19:22 | [x]

Commit my progress to GitHub | 2026-11-14T19:23 | [x]

Review my project and assignment requirements | 2026-11-15T19:23 | [x]

reuse

Since I am working solo on my 3D space shooter game, I want to create things that I can reuse in future game development projects. Some things I can reuse are my C# scripts for player movement, shooting, enemy behavior, health, and scoring. I can also reuse my Unity project structure and some of the 3D models and effects I create. The code and assets I make myself will be my original work, but if I use anything from the Unity Asset Store or other websites, I will make sure to check the license and give credit when required. I will keep track of where my assets come from in my GitHub README, including the creator's name, source link, and license information. This will help me stay organized and make it easier to use my work in future games without having to start everything from scratch.

boardNotes

Since I am working solo, my huddle board shows the decisions I made for my 3D space shooter game. I decided to use Unity as my game engine, C# for programming, Visual Studio for writing my code, and GitHub for keeping track of my progress. My main goal is to create a playable game where the player controls a spaceship, shoots enemies, and earns points. I also decided to use the ADD framework to help me focus on the most important features first.

Since I don't have any teammates, there were no team disagreements. One decision I changed was switching my game from 2D to 3D because I wanted to take advantage of Unity's 3D features and challenge myself.

Some questions I still have are how detailed my 3D models should be, how many enemies I should include, and whether I will have enough time to add extra features like different weapons or levels. For now, I plan to focus on making the basic gameplay work before adding anything more complicated. I will keep my decisions and progress recorded in my GitHub README so I can look back at them while developing my game.

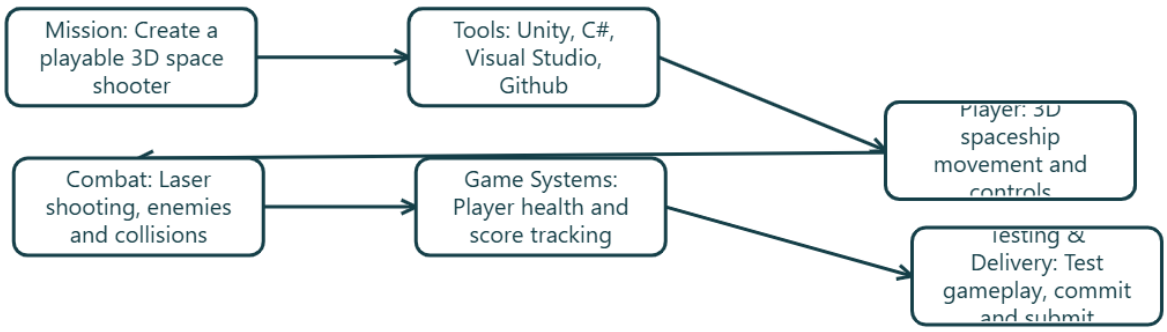
Whiteboard

My huddle board shows my plan for creating a 3D space shooter game in Unity. Since I am working solo, I am responsible for every part of the project. My mission is to create a playable game where the player controls a spaceship, shoots enemy ships, and earns points.

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I will use Unity for the game engine, C# for programming, Visual Studio for writing my scripts, and GitHub to track my progress. I will start by creating the spaceship and its movement controls. After that, I will work on shooting, enemies, collisions, health, and scoring. I will test each feature before moving on to the next one.

My main priority is to make sure the basic gameplay works before adding extra features. I will record important decisions in my GitHub README and use commits to show my progress. Since I am working solo, there are no team disagreements, but I will review my decisions and make changes when needed.



SOLO PROJECT

Whiteboard

My production pipeline shows how I plan to create my 3D space shooter game using Unity and C#. Since I am working solo, I will be responsible for every stage of development. My goal is to create a playable game where the player controls a spaceship and shoots enemy ships. I will use Unity, Visual Studio, C#, and 3D models to build my project. My main tasks will be creating spaceship movement, shooting mechanics, enemies, and collisions. After completing these features, I will test the game to make sure everything works correctly. I will also use GitHub to save my progress. This plan will help me stay organized and complete my project one step at a time.



codeTables

0

id

dc99a4d2-b60a-4f3d-833b-8b5f3552562b

title

Classes & components

columns

0

label

Class

type

text

1

label

Responsibility

type

text

2

label

GameObject

type

text

rows

0

0

PlayerController

1

MonoBehaviour, Rigidbody

2

Controls spaceship movement

1

0

PlayerShooting

1

MonoBehaviour

2

Fires laser projectiles

2

0

Laser

1

MonoBehaviour, Collider

2

Moves lasers and detects hits

3

0

EnemyController

1

MonoBehaviour, Collider

2

Controls enemy behavior

4

0

GameManager

1

MonoBehaviour

2

Tracks score and game state

5

0

PlayerHealth

1

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MonoBehaviour

2

Tracks player health and damage

1

id

abebc130-e60e-4b89-9d53-1d1db2d73a53

title

Variables / properties

columns

0

label

Name : type

type

text

1

label

Value / purpose

type

text

2

label

Used by

type

text

rows

0

0

moveSpeed

1

float

2

Spaceship movement speed

1

0

laserSpeed

1

float

2

Laser projectile speed

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2

0

health

1

int

2

Player health

3

0

score

1

int

2

Player's points

4

0

damage

1

int

2

Damage from lasers

5

0

laserPrefab

1

GameObject

2

Laser object to spawn

2

id

bc70c9c9-89ee-435e-9e43-6768e3aad06a

title

f(x) Functions

columns

0

label

Function

type

text

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1

label

Inputs result

type

text

2

label

When called

type

text

rows

0

0

MovePlayer()

1

PlayerController

2

Moves the spaceship

1

0

Shoot()

1

PlayerShooting

2

Creates laser projectiles

2

0

TakeDamage()

1

PlayerHealth

2

Reduces player health

3

0

OnTriggerEnter()

1

Laser

2

Detects laser collisions

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4

0

AddScore()

1

GameManager

2

Adds points for destroyed enemies

3

id

c4ce91e3-3214-4c93-a2a3-797e589120e0

title

Packages / using

columns

0

label

Package / namespace

type

text

1

label

Why included

type

text

2

label

Tested

type

checkbox

rows

0

0

using UnityEngine;

1

Accesses Unity classes such as MonoBehaviour, Rigidbody, and GameObject

2

false

1

0

using UnityEngine.InputSystem;

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1

Allows me to use Unity's newer Input System for spaceship controls

2

false

2

0

using UnityEngine.UI;

1

Allows me to create UI elements such as health bars

2

false

3

0

using TMPro;

1

Allows me to display score and other text using TextMeshPro

2

false

4

0

Unity Input System package

1

Handles keyboard and controller input

2

false

5

0

Unity 3D Physics

1

Handles collisions and Rigidbody physics

2

false

communicationTables

0

id

a0b87b42-1615-4ce9-a50b-36346630578e

title

Team check-ins

columns

CS110 / INDEPENDENCE ROAD

0

label

Description / contact / owner

type

text

1

label

When Pacific

type

datetime-local

2

label

Agreed

type

checkbox

rows

0

0

Review my Unity project goals

1

2026-11-11T19:21

2

true

1

0

Check my C# code and progress

1

2026-11-12T19:22

2

true

2

0

Test spaceship movement and shooting

1

2026-11-13T19:22

2

true

3

0

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Commit my progress to GitHub

1

2026-11-14T19:23

2

true

4

0

Review my project and assignment requirements

1

2026-11-15T19:23

2

true

Task list

0

id

c5e11bf3-b3a0-4d88-982a-dd63ba902650

task

Project Documentation

Task owner reference

Rodolfo Sierra-Trujillo

Deadline

2026-11-12T20:30

Completed

false

Execution / result

1

id

aff5deb9-7c82-45b2-8d3c-2a5601ef059b

task

Unity Project Setup

Task owner reference

Rodolfo Sierra-Trujillo

Deadline

2026-11-12T20:30

Completed

false

Execution / result

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2

id

44fbc2e3-60e2-4d87-ba78-f289ee084440

task

Import 3D Player Spaceship

Task owner reference

Rodolfo Sierra-Trujillo

Deadline

2026-11-12T20:30

Completed

false

Execution / result

3

id

baf10c63-9ac5-43b9-ad80-5b505fa5b54d

task

Create Player Movement

Task owner reference

Rodolfo Sierra-Trujillo

Deadline

2026-11-12T20:30

Completed

false

Execution / result

4

id

ec3b58d6-c424-458b-9dbb-f35e138edb93

task

Test Player Controls

Task owner reference

Rodolfo Sierra-Trujillo

Deadline

2026-11-12T20:30

Completed

false

Execution / result

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5

id

01e4d92d-6de5-49f6-9093-478c1c7fc284

task

GitHub Update and Handoff

Task owner reference

Rodolfo Sierra-Trujillo

Deadline

2026-11-12T20:30

Completed

false

Execution / result

revision

1